

Brendan Tomoschuk

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Skills

Data Analysis	Topic modeling, sentiment analysis, word embeddings, spaCy, NLTK, NumPy, SciPy, sklearn Classification techniques (Regression, random forest, gradient boosting), Mixed Effect modeling
Programming	Python, R, SQL (PostgreSQL), Spark, JMP

Experience

PhD Researcher 2014-Present

- Published NLP paper on predicting errors in early learners of French, Spanish and English to 87% accuracy
- Published methods paper modeling language skills from 1500 surveyed bilinguals to 91% accuracy with multivariate model
- Led three-person team to design and test novel word learning methods that improves second language word learning by up to 11% (publication in preparation)
- Directed weekly statistics and research methods classes to 30 undergraduate students

Fulbright Scholar 2016-2017

- Organized and led collaboration between Language Production Lab at UC San Diego and the LEMMA grant project at Ghent University, Belgium
- Found significant differences in language processing strategies between American and Belgian populations that inform language theory and pedagogy (publication in preparation)
- Presented findings at two international conference and to the acting American ambassador to Belgium

Analytics Projects

Reddit Sentiment and Topic Analysis <https://goo.gl/Lz85G4> 2017

- Implemented NLP pipeline to scrape, clean and process comments from Reddit television discussion threads (spaCy, sklearn)
- Analyzed topics and sentiment of threads from specific fan and general TV communities
- Found that fan comments are more likely to be positive than general viewers, and that they focus on more specific topics

Language Learning Difficulty <https://goo.gl/ipnN9S> 2017

- Modeled word-level learning difficulty of different languages from Duolingo user data with R
- Showed that French learners need more practice to achieve same proficiency of other learners and that this is driven by particular parts of speech

Education

Ph.D. Cognitive Psychology (Psycholinguistics), *University of California, San Diego* 2014-2019

B. Phil. Neurolinguistics, B.S. Science, *Pennsylvania State University* 2010-2014

Awards & Funding

- Fulbright Fellowship 2016-2017
- National Defense Science and Engineering Graduate (NDSEG) Fellowship 2015-2019